

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458336.8478 +/- 0.0052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.045
Transit depth (Rp/Rs)^2	1.9 +/- 1.24 [%]
Orbital Inclination (inc)	80.04 +/- 2.97
Airmass coefficient 1 (a1)	0.907 +/- 0.045
Airmass coefficient 2 (a2)	0.08 +/- 0.064
Scatter in the residuals of the lightcurve fit is	4.97 %
Best Comparison Star	#2 - [119, 87]
Optimal Aperture	6.24
Optimal Annulus	11.97
Transit Duration (day)	0.013 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.138 $\pm$ 0.045 vs 0.1326 (deviation 0.12 σ)
Duration fit vs published	0.013 d vs 0.0737 d (deviation 2.34 σ)
Mid-time O-C	-5.4 min
Depth SNR	3.1
Residual scatter	4.97 %

Light curve

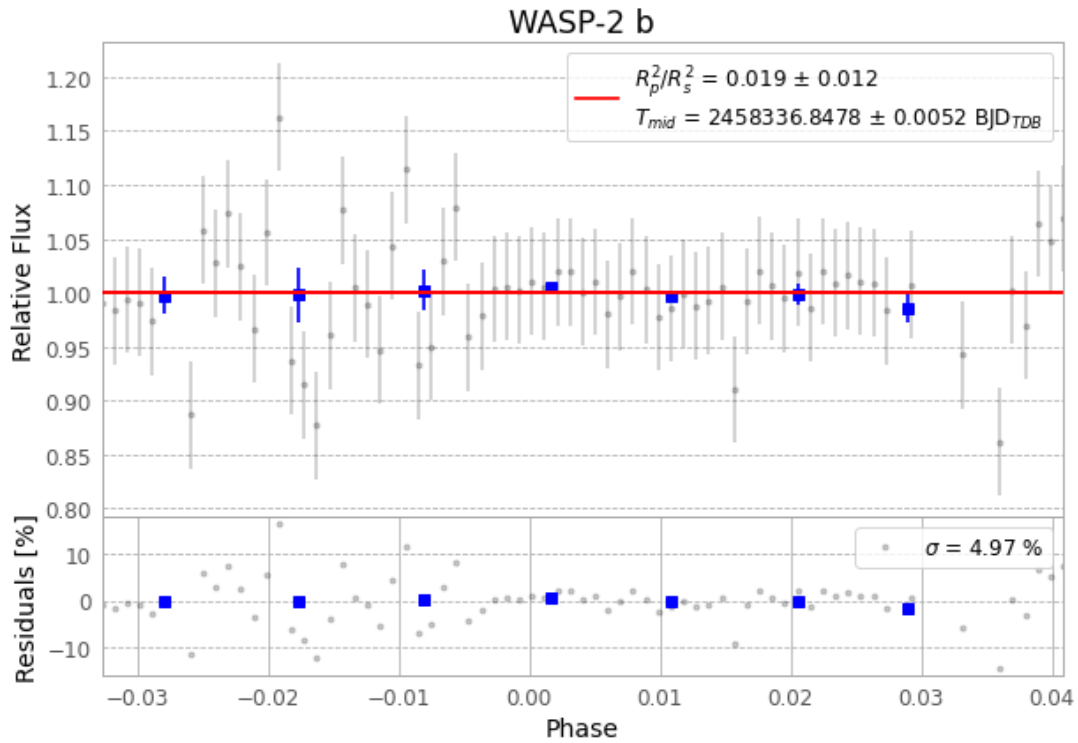


Figure 1 — FinalLightCurve_WASP-2 b_2018-08-05.png (SHA-256 6c6405427f8ccee1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [298, 217], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 21c280b886bfcaf7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e

Data provenance

75 FITS frames (49 MB), WASP-2180806063012.FITS → WASP-2180806101812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-180806.log (3e75697c1d40...), FinalLightCurve_WASP-2 b_2018-08-05.png (6c6405427f8c...), FinalLightCurve_WASP-2 b_2018-08-05.csv (b7817dd82165...)

Compute resources

Wall time 465.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 22:33Z.